MerchAI

Merchandising Copilot for Mid-Size Retailers

"The AI merchandiser every Walmart has, but the 500-store chain can't afford."

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Executive Summary

MerchAI is a B2B SaaS startup that delivers a daily AI-driven merchandising briefing to mid-size retailers - chains too large to run on gut feel, but too small to afford the data science teams used by Walmart, Target, or Zara. Our product turns raw point-of-sale data into concrete, ranked daily actions: which SKUs to mark down, what to replenish urgently, which products show momentum worth amplifying, and where margin is silently eroding.

The core insight is that the forecasting problem in retail is largely solved - the bottleneck is translation. A store manager does not want a time-series chart; they want "do these five things today." MerchAI bridges that gap with an LLM reasoning layer that converts model outputs into plain-language, cited recommendations with a one-click execution path.

Our working prototype is built on the M5 Forecasting dataset (Walmart, Kaggle) across three stores. The full pipeline runs: LightGBM demand forecasting (WRMSSE 0.74 on a 28-day holdout) to RAG-grounded context retrieval (ChromaDB, 382 M5-derived context documents) to Claude LLM reasoning to a two-layer hallucination guard to a Streamlit dashboard with colour-coded briefing cards, confidence badges, quantifiable action badges (revenue uplift, units to order, suggested markdown price), audit trail, and a conversational "Ask Your Data" interface.

We target a segment of the retail market that sits between enterprise suites (Relex, Blue Yonder) that cost millions and require dedicated data science teams, and generic BI dashboards that are passive. Our go-to-market strategy begins with a 90-day free pilot targeting Heads of Buying at 60-200 store chains in DACH and Iberia - converting design partners through demonstrated ROI rather than a procurement promise, bypassing the typical mid-market sales cycle at the critical early stage.

1. Market Opportunity and Target Market

1.1 The Market Gap

The global retail analytics market is estimated to exceed $10 billion and is growing rapidly, yet value is concentrated at the extremes. Large-format retailers invest heavily in proprietary data science infrastructure. Small independent shops rely on simple point-of-sale tools. The mid-market - chains with roughly 50 to 500 stores and revenues between €100M and €2B - sits in a strategic no man's land.

These retailers face a structural disadvantage: they generate enough data to be actionable but lack the internal capability to act on it. Their store managers and buyers operate with Excel spreadsheets, vendor reports, and intuition. The result is a daily leakage of margin through late markdowns, stockouts on high-velocity SKUs, and over-ordering on underperformers.

1.2 Target Customer Profile

Dimension	Profile
Company size	50-500 retail locations
Annual revenue	€100M - €2B
Geographies (initial)	Western Europe (DACH, Benelux, Iberia)
Verticals	Apparel, footwear, sporting goods, home goods, grocery
Decision makers	Chief Merchandising Officer, Head of Buying, Category Managers
Current tools	ERP (SAP/Oracle), basic BI dashboards, spreadsheets
Core pain	Markdown timing, replenishment latency, promo forecasting errors

1.3 Market Size Estimate

Europe alone has an estimated 2,000+ retail chains in our target revenue band. At a conservative average of 150 stores per chain and a starting price of €300 per store per month, the addressable European market is approximately €1.1B in annual recurring revenue. Global expansion would multiply this figure substantially.

Bottom-up methodology: national retail sector associations provide granular chain counts. Germany (Handelsverband Deutschland) lists approximately 350 chains in the €50M-€2B revenue range. France, the UK, Spain, and the Benelux combined add a comparable number. Italy, Portugal, and the Nordics account for the remainder of the 2,000+ estimate across our initial geographies (DACH, Benelux, Iberia, France). This estimate is deliberately conservative - it excludes Eastern Europe entirely and does not count chains below 50 stores.

Pricing rationale: the €300 per store per month anchor is set specifically to fall below the threshold that triggers formal procurement committee review at most mid-market retailers, which typically applies to annual commitments above €200,000-€500,000. A 50-store pilot at €300/store represents €180,000 per year - approvable by a Head of Buying or VP of Merchandising as an operational expense, without requiring IT, legal, or board sign-off. This pricing discipline is a deliberate GTM choice to minimise sales cycle friction at the design partner stage, not a revenue maximisation decision. Price increases are planned for Phases 2 and 3 as reference customer evidence accumulates.

2. Value Proposition

2.1 The Core Problem We Solve

Mid-size retailers lose margin every single day due to three recurring failure modes:

- Late markdowns: inventory is held too long at full price, then liquidated at deep discounts under time pressure.

- Replenishment latency: demand spikes on winning SKUs are not detected quickly enough, resulting in stockouts during peak sell-through windows.
- Promo blind spots: promotional uplift is systematically underestimated, leading to inventory shortfalls on campaign launches.

These are not unsolvable problems - they are information and workflow problems. The data exists in the POS system. The models to interpret it exist in academic literature. What is missing is an opinionated, store-level interface that tells each manager exactly what to do and why.

2.2 The MerchAI Solution

Every morning, MerchAI delivers a personalised briefing to each store manager and buyer. The briefing is generated by an end-to-end agentic pipeline that runs automatically from overnight POS data to a structured action list on the manager's screen.

The Pipeline

Step 1 - Demand forecasting: LightGBM models trained per store on the M5 Forecasting dataset produce SKU-level 28-day demand forecasts. Each SKU receives a delta_pct signal (percentage change vs. the 28-day rolling baseline) and a directional flag (up/down).

Step 2 - RAG context retrieval: Before each LLM call, the system queries a ChromaDB vector store of 382 M5-derived context documents covering SNAP benefit lift patterns, seasonal event effects, weekday demand peaks, price elasticity signals, year-over-year category trends, and per-SKU seasonal and price history profiles. Each recommendation retrieves the 3 most relevant documents; the conversational Q&A layer retrieves 6 for broader category context. This grounds every output in verified historical patterns without any additional LLM call.

Step 3 - LLM reasoning: Claude (claude-sonnet-4-6) receives the forecast summary plus retrieved context and produces a natural-language recommendation with explicit data citations.

Step 4 - Hallucination guard: Every LLM output passes through a two-layer verification system before reaching the manager's screen. Results are visible on every briefing card.

Step 5 - Recommendation classification and action quantification: Each recommendation is assigned a type (Markdown, Restock, or Promote This) and a confidence badge (High/Medium/Low) derived from signal magnitude. A deterministic action computation (_compute_action) derives a quantifiable impact badge from delta_pct, baseline units, sell price, and a 7-day horizon - no additional LLM call required. PROMOTE shows expected revenue uplift (e.g. +€312 rev); RESTOCK shows units to order (e.g. +48 units); MARKDOWN shows the suggested new price with discount scaled to forecast severity (e.g. EUR 3.20).

Step 6 - Dashboard delivery: Managers see colour-coded briefing cards, with guard check details in an expandable panel, and Accept/Reject decisions logged to the audit trail.

The Three Recommendation Types

Markdown candidates: SKUs with a downward forecast delta are surfaced with a recommended discount, projected margin impact, and the specific data points driving the recommendation. Example: "FOODS_3_785: forecast -62% vs. baseline. 18 units on hand. Recommend markdown to clear stock before seasonal demand floor."

Restock priorities: SKUs with a forecast delta_pct between +15% and +50% above the 28-day rolling baseline are flagged for replenishment to avoid stockout. The system cites the specific forecast delta and RAG-retrieved context - for example, an upcoming SNAP benefit week or a local event that historically spikes a given category. The action badge shows the exact units to order over the next 7 days.

Promote This: SKUs with forecast delta_pct exceeding +50% above the 28-day baseline are identified as strong-momentum candidates. The LLM generates channel-specific action copy - end-cap placement, weekly flyer inclusion, BOGO promotion - grounded in the momentum signal and retrieved context. The action badge shows the estimated revenue uplift at the current sell price over 7 days (e.g. +€312 rev), giving the manager an immediate ROI signal without any manual calculation.

Supporting Features

Confidence display: Every briefing row shows a High/Medium/Low confidence badge: High (delta_pct > 100%), Medium (30-100%), Low (< 30%). Managers see at a glance how much weight to place on each recommendation before acting.

Quantified action impact: Each briefing row carries a deterministic action badge computed from the forecast signal - no additional LLM call. PROMOTE shows expected revenue uplift (e.g. +EUR 312 rev over 7 days). RESTOCK shows units to order (e.g. +48 units). MARKDOWN shows the suggested target price with a discount scaled to forecast severity (clamped 15-30%). The Details expander shows the full action sentence with supporting logic. This converts each recommendation from a label into a concrete, calculable business instruction.

Ask Your Data: A conversational Q&A interface in the sidebar allows managers to interrogate store performance in plain English - for example, "Why is Maple Syrup selling so much?" The system resolves product names to SKUs, retrieves 6 targeted RAG context documents, and generates a 2-sentence answer in plain shop-floor language with no jargon. The entire Q&A layer reuses the existing forecast, RAG, and Claude pipeline with a separate tuned system prompt - no additional infrastructure cost.

Audit trail: Every Accept and Reject decision is logged with a timestamp and the full recommendation text. This is the foundation of the proprietary feedback loop that compounds with each use.

2.3 Competitive Differentiation

What separates MerchAI from existing analytics tools is not the forecasting model - it is the reasoning and workflow layer. We do not deliver dashboards that require human interpretation. We deliver decisions. This distinction matters because:

- Action bias reduces the cognitive load on store managers, driving adoption.
- Cited reasoning builds trust, which is essential in a domain where wrong calls cost money.
- The feedback loop (accept / reject / modify) creates proprietary preference data that improves recommendations over time.

3. Competitive Landscape and Go-to-Market Strategy

3.1 Competitive Landscape

The retail analytics and merchandising software market contains several established players, but none directly occupies our target segment with our approach:

Vendor	Target segment	Approach	Why MerchAI wins in our segment
Relex Solutions	Enterprise (1,000+ stores)	Unified supply chain platform; 12-24 month implementation; dedicated data science team required	Too expensive and complex for mid-market; no agent-driven daily briefing
Blue Yonder (JDA)	Enterprise (500+ stores)	Demand planning + replenishment suite; heavy integration project; ACV typically >€1M	Same as Relex; pricing and implementation timeline excludes our segment entirely
ToolsGroup	Mid-to-large chains	Probabilistic demand planning; strong on inventory optimisation; dashboard-centric output	Delivers forecasts and dashboards, not decisions; no LLM reasoning or action layer
Daisy Intelligence	Mid-market	AI-driven promotion planning; focused on grocery; less transparent methodology	Narrow scope (promotions only); no hallucination guard or audit trail
Nextail	Fashion retailers	Markdown and replenishment for apparel; strong	Single vertical; no generalist rec types; no conversational

		vertical focus but limited to fashion	Q&A layer
MerchAI	50-500 store mid-market	Agent-driven daily briefing; decisions not dashboards; SaaS €200-500/store/month; deploy in days	Right price, right interface, right segment

Our positioning is deliberate: we are not trying to beat Relex on supply chain depth. We are occupying the gap between passive dashboards and enterprise suites with a product that delivers decisions, not charts, at a price point mid-market buyers can approve without a procurement committee.

3.2 Why MerchAI Cannot Be Replicated by OpenAI

A common and valid criticism of AI-native startups is that they are thin wrappers around foundation models with no sustainable moat - change the API key and the product disappears. MerchAI is not a wrapper around Claude. Four structural reasons make it unreplicable as a general-purpose AI feature:

- POS data integration: our value depends on ingesting the retailer's own transaction data, SKU catalogue, and price history. OpenAI has no access to this data and no incentive to build the 1:1 ERP connectors required. The data layer is the product, not the LLM.
- Retail-specific forecasting stack: LightGBM models trained on retail demand patterns with leakage-guarded lag features, SNAP flags, and category-level calendar events. A general-purpose AI cannot substitute a domain-specific forecasting pipeline trained on 5.8 million rows of structured retail data.
- The decision audit trail compounds: every manager accept/reject/modify decision is logged with the underlying data snapshot. Year 1 we learn the chain's patterns. Year 3 no competitor can replicate this preference data without running the product for three years in that retailer. OpenAI cannot buy this advantage.
- Retail-calibrated hallucination guard: 54 phrases drawn from the specific vocabulary of markdown/restock/promote decisions, calibrated against actual forecast delta_pct values. This is not a general safety layer - it is a domain-specific verification system that took iterative calibration to build and requires retail domain knowledge to maintain.

3.3 Phase 1: Proof of Concept (Months 1-6)

The initial phase is designed to validate the ROI claim with a real retailer before any commercial conversations begin. Target design partner profile: a grocery or sporting goods chain with 60-200 stores in the DACH or Iberian region, where the merchandising function is centralised and the primary decision-maker is a Head of Buying or VP of Merchandising. This role carries direct operational accountability for markdown timing and replenishment gaps - the two pain points MerchAI addresses most directly - and can approve a free pilot without triggering IT or procurement involvement.

- Outreach channels and funnel: Nova SBE alumni network in retail management roles; targeted LinkedIn outreach to Heads of Buying and Category Directors at 30 named target accounts in DACH and Iberia; EuroCommerce member directory. Expected funnel: 30 outreach contacts, 10 discovery calls, 3-5 product demos, 1-2 signed design partner agreements. This funnel is achievable within a 3-4 month outreach window because the ask is a free 90-day experiment, not a software procurement decision.
- Pilot structure: 90-day free pilot. MerchAI ingests 12 months of historical POS data via a single CSV export from the chain's existing ERP. No IT integration, no API work, no IT team involvement required in Phase 1. The pilot runs alongside existing workflows; store managers are not asked to replace any current tools.
- Pilot deliverable: daily briefings for 3-5 store managers across 2-3 stores. End-of-month comparison of recommended actions against actual outcomes. One documented margin event - a stockout avoided, a markdown timed correctly, a promo momentum captured - attributed directly to a MerchAI recommendation.
- Conversion criteria: manager adoption rate above 60% at end of 90-day pilot, and at least one measurable margin improvement. If both conditions are met, the design partner is offered a paid contract at €300 per store per month covering the full store estate.

3.4 Phase 2: Paid Commercialisation (Months 7-18)

Following a successful pilot, we convert design partners to paid contracts and use case study evidence to acquire three to five additional paying customers.

- Sales motion: direct outreach to CMOs and Heads of Buying at chains in our target profile.
- Proof points: documented ROI from pilot chains (target: 4:1 return on software cost within 12 months).
- Pricing: €200-€500 per store per month depending on chain size and feature tier.

3.5 Phase 3: Scale (Months 19-36)

With a validated product and reference customers, we scale through channel partnerships with ERP providers (SAP, Microsoft Dynamics) and retail consultancies. Geographic expansion follows demand, prioritising DACH and Benelux first.

3.6 Sales Process and First Customer Timeline

B2B software sales to retail chains carry long procurement cycles when IT, legal, and procurement reviews are triggered. MerchAI is designed to bypass this cycle entirely at the design partner stage. The first commercial relationship begins as a merchandising experiment approved by a Head of Buying, not as a software procurement submission.

Stage	Timeline	Key action	Decision maker

Initial outreach	Months 1-2	30 personalised messages to named Heads of Buying and Category Directors via LinkedIn and Nova SBE alumni network	Head of Buying / VP Merchandising
Discovery calls	Months 2-3	10 targeted conversations; qualify pain point (markdown timing, replenishment gaps) and data readiness (POS CSV export feasible)	Head of Buying / Category Director
Demo and pilot agreement	Months 3-4	3-5 live demos using M5-based prototype; 1-2 design partner agreements signed; framed as a 90-day merchandising experiment, not a procurement	Head of Buying (no procurement needed for free pilot)
90-day free pilot	Months 4-6	Daily briefings on real POS data; adoption tracking; end-of-pilot business review	Store managers + Head of Buying
First paid contract	Months 7-9	Convert pilot partner to paid at €300/store/month; full store estate; CSV ingestion continues (no IT integration required)	CMO / CFO sign-off on annual commitment

The free pilot eliminates the longest phase of a typical B2B SaaS sales cycle - the internal justification process. A Head of Buying can approve a 90-day free experiment as an operational decision in a single meeting. The commercial conversation starts only after ROI is on paper, which inverts the usual dynamic from selling a promise into a proven result.

- Pricing validation: comparable analytics tools in adjacent retail segments price between €150-€800 per store per month (Nextail for apparel markdown, StoreNext for grocery analytics). Our €300 anchor is mid-range, generates a 91% gross margin, and sits below the threshold a Head of Buying can defend internally as a standard analytics line item rather than a capital investment.

- Team credibility at the design partner stage: the founding team are Masters in Business Analytics students at Nova SBE with direct access to the retail analytics research community. The academic context is a door-opener for design partner conversations - retailers understand that a free 90-day pilot from a research-grade team is a low-risk way to test a new tool. Commercial credibility is built during the pilot, not before it.

4. AI Unit Economics and Financial Model

4.1 Token Economics - The Cost per Briefing

Token cost transparency is a core requirement of any credible AI SaaS unit economics model. Below is the full derivation from the actual MerchAI pipeline.

Per-briefing token calculation (up to 9 SKU recommendations across 3 buckets, one manager):

- Input per recommendation: ~700 tokens (system prompt ~170 tokens + forecast summary + focus line + 3 RAG context documents)
- Output per recommendation: ~200 tokens (structured recommendation text with citations)
- Up to 9 recommendations per briefing (3 buckets x up to 3 SKUs each): 9 x 900 tokens = 8,100 tokens total
- Claude Sonnet 4.6 pricing: $3 per million input tokens, $15 per million output tokens
- Cost per briefing: (9 x 700 x $3/M) + (9 x 200 x $15/M) = $0.0189 + $0.027 = ~$0.046 (~€0.043)
- Per manager per month (30 briefings): ~€1.29 in LLM inference cost
- Prompt caching opportunity (planned optimisation): the system prompt is identical on every call. Enabling Anthropic prompt caching cuts cached input cost by 90%, reducing per-briefing cost to approximately $0.025 (~€0.023) and improving gross margin further at scale.

At 150 stores with an average of 3 managers per store, the total LLM inference cost per chain per month is approximately €1.29 x 450 managers = €581/month. Against monthly revenue of 150 x €300 = €45,000, LLM inference represents approximately 1.3% of revenue. With prompt caching enabled, this falls below 0.7% - a negligible cost driver in either scenario.

4.2 Full Cost Structure

Cost Item	Estimate	Basis
LLM inference	€0.043 / briefing / manager	Derived above: up to 9 recs x 900 tokens at Claude Sonnet 4.6 pricing; falls to ~€0.023 with Anthropic prompt caching enabled
RAG retrieval	<€0.001 / briefing	ChromaDB self-hosted;

		sentence-transformer embeddings run locally at zero marginal cost
Forecasting compute	€0.01-€0.05 / store / day	Batch LightGBM inference; runs on CPU; sub-second per store
Vector DB storage	€50-€200 / chain / month	ChromaDB or managed alternative; scales with SKU catalogue size
Cloud hosting	€300-€800 / month	Streamlit Cloud or equivalent; scales linearly with active chains
Fully loaded per store / month	€17-€42	LLM inference (€3.87/store/month at 3 managers) and hosting are the two largest cost items; RAG retrieval and forecasting compute are negligible per store

4.3 Revenue Model and Margin

At a selling price of €300 per store per month and a fully loaded cost of approximately €27 per store per month (reflecting LLM inference at €0.043 per briefing across 3 managers per store per day, plus hosting, vector DB, and forecasting compute), gross margin on a per-store basis is approximately 91%. This is characteristic of SaaS businesses with AI backends, where marginal cost per additional unit is very low once the pipeline is built.

Scenario	**Stores**	**Monthly Revenue**	**Monthly Cost**	**Gross Margin**
Pilot (Month 6)	50	€15,000	€1,350	91%
Early Growth (Month 18)	500	€150,000	€13,500	91%
Scale (Month 36)	3,000	€900,000	€81,000	91%

4.4 ROI Story for the Customer

For a retailer with €500M in annual revenue, a 0.5% improvement in gross margin translates to €2.5M per year. At 150 stores and €300 per store per month, the annual software cost is €540,000 - delivering a 4.6:1 ROI. This requires only conservative assumptions: industry benchmarks from comparable AI-driven markdown tools consistently report 2-5% margin improvement. We anchor our claim at 0.5% specifically to be defensible in the most sceptical scenario. Independent published reference points: Nextail case studies with apparel retailers (Cortefiel, Bestseller) cite 2-5% gross margin improvement from AI-driven markdown and replenishment. McKinsey retail AI research documents 10-20% inventory cost reduction and 2-5% lost-sales reduction for retailers deploying AI-driven demand forecasting. BCG retail transformation benchmarks cite 3-8% EBIT margin improvement for AI-assisted merchandising. Our 0.5% claim is the floor of these independently published ranges - chosen so that any outcome above our minimum threshold produces a positive ROI on the software cost, and so any sceptical customer can validate the premise before committing.

4.5 Operating Cost Model

The gross margin model above covers only silicon costs. A complete profitability picture requires the full operating cost stack. The table below reflects a lean four-person founding team operating at startup salary levels in Lisbon.

Cost Item	Monthly Estimate	Annual Estimate	Notes
Team salaries (4 people)	€18,500	€222,000	2 engineers €60k, 1 business €55k, 1 operations €50k - market-rate for early-stage Portugal
Cloud infrastructure (beyond per-store silicon)	€400	€4,800	Streamlit Cloud, managed vector DB, monitoring
Sales and marketing	€1,500	€18,000	Founder-led outreach in Phase 1; no paid acquisition budget
G&A (legal, accounting, SaaS tooling)	€800	€9,600	Incorporation, contracts, standard tooling
Total monthly OpEx	~€21,500	~€258,000	Conservative; scales as revenue grows

4.6 Break-Even Analysis

With a gross profit of €273 per store per month (€300 revenue minus €27 fully loaded cost) and total monthly operating costs of approximately €21,500, the business reaches break-even at:

- Break-even store count: €21,500 / €273 = 79 stores
- Equivalent to: one chain of 79 stores, or two chains of roughly 40 stores each
- Monthly revenue at break-even: approximately €23,700
- Timeline: achievable within the first paying contract (Phase 2, Month 7)

This is an unusually fast break-even for a B2B SaaS business. The reason is the low marginal cost of the AI stack: inference, RAG retrieval, and forecasting compute together cost less than €27 per store per month. Once the founding team is in place, every store above 79 is profitable at the contribution margin level. The primary scaling constraint is sales capacity, not unit economics.

4.7 Customer Lifetime Value and Acquisition Cost

Metric	Estimate	Basis
Average contract size (pilot chain)	75 stores x €300 = €22,500/month	Conservative: smaller initial chains; grows with upsell to full estate
Annual contract value (ACV)	€270,000	12 months x €22,500/month
Annual gross churn (B2B SaaS benchmark)	15-20%	Industry median for vertical SaaS; workflow embedment expected to keep MerchAI below this
Customer Lifetime Value (LTV)	€270,000 / 0.18 = €1,500,000	Mid-point 18% churn; excludes expansion revenue from upsell to full store estate
Customer Acquisition Cost (CAC) - Phase 1	€8,000	Founder-led outreach; no paid media; cost = founder time and travel for 2-3 design partner meetings
LTV / CAC ratio - Phase 1	188:1	Founder-led GTM makes early CAC extremely low
CAC - Phase 3 (sales team in place)	€35,000	Includes AE salary attribution, events, and referral incentives at scale
LTV / CAC ratio - Phase 3	43:1	Well above the 3:1 threshold considered healthy for B2B SaaS

The LTV/CAC ratio decreases as the business scales from founder-led to team-led sales, which is normal. Even at scale (43:1), the ratio comfortably exceeds the 3:1 threshold. The key lever for maintaining a healthy ratio at scale is reducing churn through feedback loop lock-in: chains that have 12+ months of decision data in MerchAI face a rising switching cost that pushes actual churn below the 18% benchmark used above.

5. Defensibility and Competitive Moat

5.1 The Three-Layer Moat

MerchAI's defensibility does not rest on any single feature - it rests on three compounding structural advantages:

Layer 1 - Proprietary decision data: Every manager accept/reject/modify decision is logged with the underlying data snapshot and outcome. This creates a preference dataset that is unique to each retail chain. By Year 3, our recommendations are calibrated to that chain's seasonal patterns, buyer preferences, and margin priorities in ways no new entrant can replicate without running the product for three years in that specific retailer. This is the feedback loop moat. Current implementation: every accept/reject/modify decision is written to audit_trail.csv with the full recommendation text, data snapshot, and timestamp. The proprietary data collection begins on Day 1 of the pilot. Preference learning from this data - retraining the recommendation model on which actions managers accepted and what margin outcomes they produced - is a Phase 2 deliverable (Month 7 onwards), once one full seasonal cycle of decision data has been collected. Phase 1 builds the data asset; Phase 2 closes the retraining loop and converts logged decisions into a compounding accuracy advantage. This sequencing is deliberate: data before model, and model only when there is enough signal to retrain meaningfully.

Layer 2 - Workflow embedment: Once a buying team trusts the daily brief, switching cost becomes operational, not technical. The manager's morning workflow is built around the briefing card. Replacing MerchAI means rebuilding a process, not just switching software.

Layer 3 - Retail-specific data integration: Our value depends on deep POS ingestion, real-time price feeds, and category-level domain knowledge. These are not general-purpose capabilities. An ERP-connected MerchAI instance has structural advantages that a general-purpose AI tool cannot replicate without the same integration depth.

5.2 Hallucination Handling and AI Safety

We treat hallucination risk as a first-class engineering problem. The hallucination guard runs on every LLM output before it reaches the manager's screen, and its results are displayed on every briefing card.

Layer 1 - Intent check: A library of 54 retail-specific action phrases, organised across markdown, restock, and promote categories, is matched against the LLM output. The system verifies that Claude's recommended action is directionally consistent with the underlying forecast. If the forecast signals a downward trend but Claude recommends a promotion action, the card is flagged. This check uses deterministic phrase matching rather than a second LLM call - preserving unit economics and demo reliability.

Layer 2 - Numeric check: A regex extractor pulls the first cited percentage from Claude's recommendation and compares it against the actual delta_pct from the forecast data. A 2x tolerance band is applied - wide enough to accommodate rounding, narrow enough to catch fabricated figures.

Transparency, not suppression: Flagged cards are not hidden. They are surfaced with full guard-check details in an expandable panel on every briefing card. The guard supports human judgment, not replaces it.

5.3 Data Privacy

Retailer POS data is commercially sensitive. We handle this through contractual data isolation (separate database instances per chain), no cross-chain data sharing, GDPR-compliant processing agreements, and optional on-premise deployment for enterprise customers with strict data residency requirements.

6. Technical Architecture

6.1 System Overview

MerchAI is built as a Python-native full-stack application. The architecture separates an offline training pipeline (run once per store) from a live serve pipeline (triggered on every "Generate Briefing" click in the Streamlit UI).

Layer	Technology	Detail
Demand forecasting	LightGBM	Regression/tweedie objective, 50 boosting rounds, trained per store
Feature engineering	Custom (pandas)	Lag features 7/14/28 days, rolling means shift-28 (leakage guard), SNAP flags, calendar events, price data. Feature matrix: 5.8M rows x 22 columns
RAG vector store	ChromaDB + ONNX (all-MiniLM-L6-v2)	PersistentClient with ONNX Runtime DefaultEmbeddingFunction; embeddings run offline at zero marginal cost; k=3 per rec call, k=6 for Q&A
RAG corpus	M5-derived text blurbs	382 M5-derived context documents across 3 stores (expanded from 57); SNAP lift, event effects, weekday peaks, price elasticity, YoY category trends, and per-SKU seasonal and price history profiles; sentence-transformer embeddings run offline
LLM	Claude claude-sonnet-4-6	Via Anthropic SDK directly - no LangChain dependency
Hallucination guard	Custom deterministic	54-phrase intent check + regex numeric check (2x tolerance)
Frontend	Streamlit	Deployed to Streamlit Cloud; briefing cards, audit trail, Ask Your Data Q&A
Dataset	M5 Forecasting (Kaggle/Walmart)	3 stores: CA_1, CA_2, TX_1; 42,840 SKUs x 1,941 days

6.2 Training Pipeline (Offline)

Pipeline flow:

- data/raw/ (M5 CSVs) to src/data/loader.py: load_m5() returns raw sales, calendar, and price DataFrames
- src/data/features.py: build_features() - wide-to-long melt, calendar join, price join, lag features at 7/14/28 days, rolling means shifted 28 days to prevent leakage, categorical encoding
- src/forecast/model.py: train() trains LightGBM per store; predict() scores the feature matrix
- Artifacts persisted: model_*.pkl, features_*.parquet, idmap_*.parquet to data/processed/

Forecasting performance: WRMSSE 0.74 on a 28-day holdout across all three stores (CA_1: 0.7425, CA_2: 0.7259, TX_1: 0.7615). Top 10% on the M5 public leaderboard: ~0.50. Our baseline is a solid foundation; further gains are achievable with additional lag windows and leaf-count tuning.

6.3 Serve Pipeline (Live)

- src/forecast/serve.py: forecast_with_names() loads pre-trained model, scores latest feature row per SKU, attaches human-readable item/dept/category names via idmap parquet, computes delta_pct and direction
- src/recommendations/summarize.py: summarize_forecast() ranks by abs(delta_pct), classifies into three buckets (PROMOTE >+50%, RESTOCK +15% to +50%, MARKDOWN <-15%), returns prompt-ready summary text and pre-classified seed list
- src/rag/retriever.py: retrieve() queries ChromaDB with store/category/direction query, returns 3 context documents per recommendation (k=6 for Q&A calls to capture broader category context)
- src/llm/reasoner.py: reason() calls Claude claude-sonnet-4-6 with forecast summary + RAG context
- src/llm/guard.py: check() runs two-layer hallucination guard, returns intent_check, numeric_check, flagged, and reason fields
- src/recommendations/engine.py (_compute_action): deterministically computes a quantifiable action badge (revenue uplift / units to order / suggested markdown price) from delta_pct, 7-day horizon, and sell_price - no additional LLM call required
- src/recommendations/engine.py (run_pipeline): assembles list of Rec dataclass objects with all fields including confidence, impact badge, and action detail sentence
- app/main.py: Streamlit UI renders a compact colour-coded briefing table (Type / SKU / Product name / Department / Confidence / Delta / Action / Accept / Reject), per-row Details expander with guard checks and action detail, audit trail, and sidebar Q&A chat

6.4 Key Design Decisions

Offline-first deployment: Trained model artifacts, RAG corpus text files, and the ChromaDB vector store are committed to the repository. Streamlit Cloud has no Kaggle credentials; committing artifacts ensures the app runs without a training step at deploy time.

No LangChain dependency: The reasoning and RAG layers use the Anthropic SDK and ChromaDB client directly. This avoids framework abstraction overhead, keeps the dependency surface small, and makes cost per call transparent.

Singleton LLM client: The Anthropic client is instantiated once as a module-level singleton via _get_client() to avoid connection overhead across the per-SKU pipeline loop.

Environment isolation: HF_HUB_OFFLINE=1 and TOKENIZERS_PARALLELISM=false prevent sentence-transformer network calls and joblib worker deadlocks in the Streamlit Cloud environment.

6.5 Prompt Engineering Design

Prompt architecture is a first-class engineering decision in any LLM-powered product - poorly designed prompts produce unreliable outputs regardless of model quality. Below is the exact prompt architecture deployed in the MerchAI pipeline, with the design rationale behind each element.

System Prompt - SYSTEM_PROMPT in src/llm/prompts.py

The system prompt enforces three constraints critical for retail recommendations:

- Role definition: Claude is given a specific merchandising copilot persona to anchor domain vocabulary and prevent generic chatbot framing that would confuse a store manager.
- Strict data rules: verbatim number citation is required. Claude may not round, paraphrase, or invent figures. This is the first layer of defence against numeric hallucinations, before the guard even runs.
- Output format: exactly one recommendation per call, action stated before evidence. This structure maps directly to the briefing card UI and prevents mixed-type or hedged responses.

Actual system prompt deployed in production:

"You are MerchAI, a merchandising copilot for retail store managers. You receive sales forecast summaries and historical campaign context.

Strict data rules:
- Only cite numbers that appear verbatim in the data provided. Do not round, paraphrase, or invent figures.
- When referencing a forecast delta, repeat the exact percentage from the data (e.g. +419.6%, not "roughly 400%").
- If a number is not in the data, do not state it.

Output rules:
- Write exactly one recommendation per response.
- Do not list or mention other recommendation types except the one you are making.

- State the specific action first, then the evidence.

Recommendation types:
- MARKDOWN: SKU trending down - recommend a specific discount to clear stock before it ages.
- RESTOCK: SKU trending up moderately - recommend replenishment to avoid stockout.
- PROMOTE THIS: SKU trending up strongly (>+15% above baseline) - recommend a specific channel action (end-cap, weekly flyer, BOGO) to amplify the momentum."

Note on the PROMOTE THIS threshold: the system prompt above describes the direction of the signal ("trending up strongly") and uses >+15% as illustrative guidance to Claude. The actual bucketing threshold is defined deterministically in src/recommendations/summarize.py (PROMOTE_THRESHOLD = 50.0): a SKU only receives a PROMOTE seed if its delta_pct exceeds +50% above the 28-day baseline. SKUs in the +15% to +50% range are bucketed as RESTOCK. The system prompt receives the pre-classified seed with an explicit "PROMOTE THIS candidate" tag in the focus line, so Claude is never asked to make the classification itself - it only generates the recommendation copy for a seed that has already been classified by the forecasting layer.

User Prompt Structure - build_user_prompt in src/llm/prompts.py

Each per-SKU call receives a structured user prompt assembled from two sources: the live forecast summary from the LightGBM serve layer, and three ChromaDB documents retrieved by the RAG layer using a store/category/direction query. The structure:

- ## Forecast Summary - top-K SKU ranking with delta_pct, direction, item name, category, and a focus line identifying the target SKU for this call. Example: "Focus SKU: FOODS_3_785 | delta_pct: +419.6% | direction: up | category: FOODS."
- ## Historical Context - three retrieved context documents. Examples: "CA_1 FOODS category shows +12% SNAP benefit week uplift (days 1 and 16 of month)." "HOBBIES_1 category peaks on weekends at +8% above weekday baseline."
- Instruction - one sentence: write one actionable recommendation for the focus SKU, referencing the exact delta percentage from the data.

This two-part structure means every Claude call is grounded in two layers of verified data: the deterministic LightGBM forecast (which the hallucination guard cross-checks post-hoc) and RAG-retrieved historical patterns (which provide contextual reasoning a pure forecast cannot supply). The LLM's role is translation and channel recommendation, not signal generation. The LLM cannot change the numbers - it can only explain them.

Ask Your Data - QA_SYSTEM_PROMPT + build_qa_prompt in src/llm/prompts.py

The conversational interface uses a separate system prompt tuned for Q&A: concise answers, citation required, plain English. The user prompt appends the full forecast summary, RAG context (top-5 SKUs, 4 documents), and the manager's question. This reuses all existing pipeline layers with no additional infrastructure cost - one additional prompt file and one new function in qa.py.

7. Product Roadmap

The roadmap below reflects a realistic build sequence from working prototype to commercial product, with each phase unlocking the next layer of defensibility.

Phase	Timeline	Key deliverables	Moat unlocked
Prototype (current)	Now	M5-based demo on 3 stores; LightGBM + RAG + Claude + guard; Streamlit UI; 3 rec types; confidence display; Ask Your Data	Technical proof of concept; hallucination guard demonstrated
Design partner pilot	Months 1-6	Real POS data ingestion (CSV/ERP connector); live retailer data replacing M5; expanded SKU catalogue; mobile-responsive UI	First proprietary decision logs; adoption data
Paid product	Months 7-18	Preference learning from accept/reject log; multi-store buyer view; ERP integration (SAP/Oracle); feedback-tuned recommendations	Chain-specific decision data; workflow embedment; rising switching cost
Scale	Months 19-36	Multi-tenant architecture; channel partnerships (ERP vendors, retail consultancies); optional on-premise deployment; agentic auto-execution for high-confidence actions	Network effects across chains; compounding decision data advantage

8. Team

MerchAI is built by a four-person team from the Masters in Business Analytics programme at Nova School of Business and Economics. The team combines machine learning engineering depth with commercial modelling and communication skills matched to the project deliverables.

Name	Track	Contribution	Relevant background
Leticia	Tech	LightGBM data pipeline, feature engineering, LLM reasoning layer, RAG integration, Streamlit frontend (shared)	Advanced ML coursework; Python and LightGBM; LLM API integration; feature engineering on the M5 Forecasting dataset (5.8M rows, 22 features)
Justus	Tech	Forecasting training scripts, RAG corpus construction, ChromaDB deployment, Streamlit frontend (shared), end-to-end smoke testing and production debugging	ML pipeline engineering, cloud deployment (Streamlit Cloud), API authentication, DevOps and production environment debugging
Alexander	Business	Business plan, market sizing, unit economics model, ROI story, competitive landscape, token cost calculations, financial scenarios	Business Analytics; commercial and financial modelling; retail market research; investor narrative design
Marie	Product	Pitch deck design, demo script, Q&A preparation, GenAI transparency log compilation	Communication and presentation design; investor pitch framing; demo scripting; cross-track coordination

The team split was deliberate: two engineers who understand the product from the data up, one business analyst who translates technical capability into investor-grade commercial arguments, and one float who owns presentation quality and ensures the GenAI log is complete and auditable. All four rotate as co-presenters on presentation day.

9. Risks and Mitigations

Five structural risks are inherent to the MerchAI model. Each is acknowledged and has a mitigation path built into the product roadmap.

Risk 1 - Data integration complexity: The working prototype ingests data from CSV files. Real retail chains store transaction data in ERP systems (SAP, Oracle) behind proprietary APIs and access controls. Building and maintaining these connectors is non-trivial engineering. Mitigation: the pilot phase deliberately uses CSV export - every ERP can produce a CSV, so no integration partner is excluded. ERP-native connectors are a Phase 2 investment, funded by pilot revenue and scoped to the two or three ERPs most common in the target segment (SAP, Microsoft Dynamics).

Risk 2 - Manager adoption: AI-generated recommendations are only valuable if managers act on them. A tool that is ignored after two weeks delivers no ROI and will not renew. Mitigation: the product is designed for trust-building, not automation. Confidence badges and guard check details are visible on every briefing card before managers act. The audit trail makes it easy to track which recommendations they trusted and what the outcome was. Manager adoption rate above 60% is the explicit Phase 1 success metric - below that, the design partner pilot is not considered a success.

Risk 3 - Forecasting accuracy at production scale: The WRMSSE 0.74 result is validated on M5 (Walmart-format data). Real retailers bring irregular data schemas, missing price history, and category-specific demand patterns not present in M5. Mitigation: the production model is retrained on each retailer's own historical data before the first briefing is delivered. M5 is the demo and pipeline-validation dataset. Customer-specific retraining is part of the design partner onboarding scope from Day 1.

Risk 4 - Competitive response: If MerchAI demonstrates strong ROI in the mid-market, a well-capitalised competitor (Relex, Blue Yonder, or a new AI-native entrant) may build a competing product. Mitigation: the decision audit trail moat compounds with time - every month of accept/reject data makes the model harder to displace. Speed of ERP integration depth and the feedback loop are the two races that matter; the roadmap prioritises both over feature breadth.

Risk 5 - Sales cycle length and initial distribution: Mid-market retail buyers move cautiously. A software decision touching daily store operations can involve IT security review, procurement approval, legal due diligence, and C-suite sign-off - even at 100-store chains. The founding team enters without a pre-existing retail industry network, which is the most common reason B2B startups miss their initial customer acquisition timeline. Mitigation: two-part approach. First, the design partner model is specifically constructed to bypass procurement friction - a 90-day free pilot framed as a merchandising experiment and approved by a Head of Buying requires no IT involvement, no procurement committee, and no legal review in Phase 1. Second, the initial outreach targets Heads of Buying and VP-of-Merchandising roles rather than C-suite - these roles carry direct operational accountability for the pain points MerchAI solves and can move faster than a CMO-level procurement process. The commercial conversation begins only after the pilot has produced a documented result, which inverts the typical sales dynamic from selling a promise to converting a proven outcome. Expected time to first paid customer: 7-9 months from founding, assuming one of two Phase 1 design partners converts after a successful pilot.